Free Space Optical Communications and DataComm calculations.

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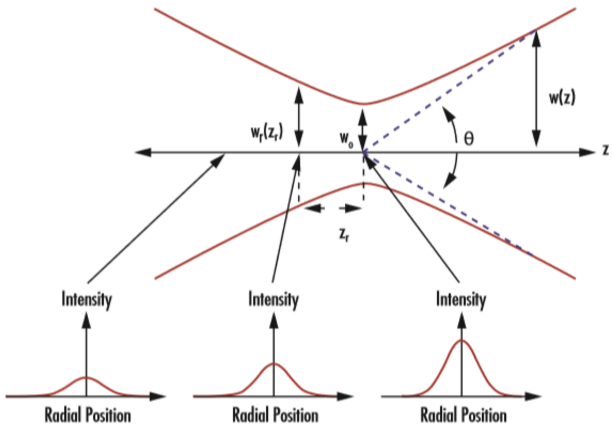
Contents

1 Laser beam propagation

Most equations according to Andrew-Phillips book.



In the plot, $W_r(z_r)$ belongs to beam width at Rayleigh distance z_r, W_0 to the beam waist and $W(z)$ to the beam width at any distance z, and θ the beam divergence.

1.1 Divergence

$$\theta = \frac{\lambda}{\pi W_0}$$

1.2 Spot size

With $W_0 = w_0$, the beam spot size is given by:

$$w(z) \approx w_0 + z\theta$$

with differs from beam diameter, which is $2w(z)$ using this convention referred to the graphic.

1.3 Field of View (FOV)

$$\mathrm{FOV} = 2\alpha = 2\tan^{-1}\left(\frac{d}{2f}\right) \approx \frac{d}{f}$$

1.4 Gaussian beam

$$U_0(r, 0) = \exp\left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0}\right)$$

$$U_0(r, L) = \frac{1}{p(L)} \exp(ikL) \exp\left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0}\right)$$

$$p(L) = 1 - \frac{L}{F_0} + i\frac{2L}{kW_0^2}$$

1.4.1 Gaussian beam parameters

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

1.4.2 Gaussian Beam width

$$W = W_0\sqrt{\Theta_0^2 + \Lambda_0^2}$$

1.4.3 Beam width at FWHM and $1/e^2$

$$W_{1/\mathrm{e}^2} = \frac{W_{\mathrm{FWHM}}}{\sqrt{2\ln 2}}$$

1.4.4 Divergence at FWHM and $1/e^2$

$$\theta_{1/\mathrm{e}^2} = \frac{2\lambda}{\pi W_0(0)}$$

$$\theta_{\mathrm{FWHM}} = \sqrt{\frac{\ln 2}{2}} \cdot \theta_{1/\mathrm{e}^2}$$

1.4.5 Irradiance on-axis

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2}\left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

2 Link budget

2.1 Power at the receiver

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{W^2}}\right) [\mathrm{W}]$$

2.2 Irradiance at the receiver

$$I_{R_x} = 4\frac{P_{R_x}}{\pi D_{R_x}^2}\left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

	Component	Symbol	Downlink	Uplink	Unit
Tx	Mean power	P_{Tx}			dB
	Gain	G_{Tx}			dB
	Transmission loss	L_{Tx}			dB
	Pointing loss	L_{PEB}			dB
Channel	Range loss				dB
	Atmospheric attenuation	L_{atm}			dB
	Scintillation loss	L_{sci}			dB
Rx	Gain	G_{Rx}			dB
	Reception loss	L_{Rx}			dB
	Splitting loss				dB
Budget	Power at detector	P_{Rx}			dB
	Required power				dB
	Link margin				dB

3 Atmospheric channel effects

3.1 Kolmogorov outer scale and inner scale

Inertial range:

$$D_{RR}(R) = \left\langle (V_1 - V_2)^2 \right\rangle = C_V^2 R^{2/3}$$

$$D_{RR}(R) = \begin{cases} C_v^2 \ell_0^{-4/3} R^2 & , \ 0 < R << \ell_0 \\ C_v^2 R^{2/3} & , \ \ell_0 << R << L_0 \end{cases}$$

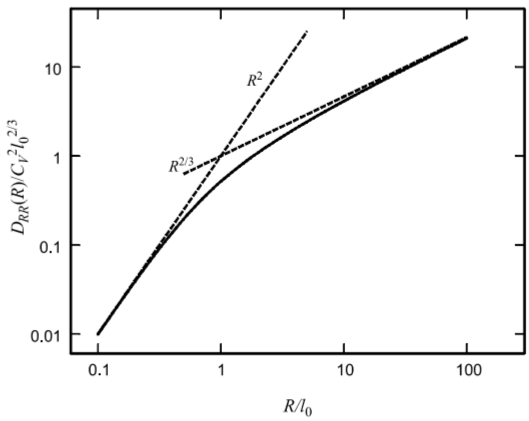


Figure 3.2 The scaled longitudinal velocity structure function showing both R^2 and $R^{2/3}$ asymptotic domains (dashed lines).

3.2 Beam spreading (Spot size)

Long-term spot size W_{LT}:

$$W_{\mathrm{LT}}^2 = W^2 + W^2 T_{SS} + \left\langle r_c^2 \right\rangle = W_{\mathrm{ST}}^2 + \left\langle r_c^2 \right\rangle$$

with $\left\langle r_c^2 \right\rangle$ the beam wander displacement variance.

Loss (Uplink)jki09:

$$L_{SR} = \left[1 + \left(\frac{D_T}{r_0}\right)^{\frac{5}{3}}\right]^{-\frac{6}{5}}$$

3.3 Beam wander

Beam wander variance:

$$\langle r_c^2 \rangle = 0.54\,(H - h_{OGS})^2 \sec^2(\zeta) \left(\frac{\lambda}{2W_0}\right)^2 \left(\frac{2W_0}{r_0}\right)^{5/3}$$

RMS beam wander displacement:

$$\sqrt{\langle r_c^2 \rangle} = 0.69 \left(\frac{\lambda}{2W_0}\right) \left(\frac{2W_0}{r_0}\right)^{5/6}$$

with r_0 the Fried's parameter.

3.4 Refractive index structure parameter

3.5 HV: Hufnagel-Valley

$$C_n^2(h) = 0.00594 \left(\frac{w}{27}\right)^2 (10^{-5}h)^{10} \exp\left[-\frac{h}{1000}\right] + 2.7 \times 10^{-16} \exp\left[-\frac{h}{1500}\right] + A \exp\left[-\frac{h}{100}\right]$$

with $A = 1.17 \times 10^{-14}\mathrm{m}^{-2/3}$ and $w = 21$ m/s recommended by ITU for night observations.

3.6 Rytov variance

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$$

3.7 Fried parameter (Coherence width)

For C_n^2 constant, horizontal propagation with uniform turbulence:

$$r_0 = (0.16 \cdot C_n^2 k^2 L)^{-3/5}$$

For Horizontal link (plane wave):

$$r_0 = \left(1.46 C_n^2 k^2 L\right)^{-3/5}$$

For Vertical link or slant paths:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^{H} C_n^2(h) dh\right]^{-3/5}$$

with ζ the zenith angle ($\sec(0) = 1$), h_0 the height above ground level of the uplink transitter and/or downlink receiver, $H = h_0 + L\cos(\zeta)$ is the satellite altitude.

Downlink can be approximated by a plane wave. Use vertical link formula.
Uplink can be approximated by a spherical wave. Use next formula:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^{H} C_n^2(h) \left(1 - \frac{h - h_{OGS}}{H - h_{OGS}} \right) dh \right]^{-3/5}$$

with H the satellite altitude, h_{OGS} the altitude of the OGS receiver, and h the integrand.

3.8 Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^{H} C_n^2(h) V_{\text{wind}}^{5/3}(h) dh \right]^{3/5}$$

3.9 Scintillation

Scintillation index:

$$\sigma_I^2 =$$

Loss:

$$L_{SI} = 4.343 \{ \operatorname{erf}^{-1}(2\rho_{th} - 1) \left[2 \ln(\sigma_I^2 + 1) \right]^{\frac{1}{2}} - \frac{1}{2} \ln(\sigma_I^2 + 1) \}$$

4 Link Distance

With R_E the Earth radius, H_{OGS} the ground station height, H or H_{sat} the satellite altitude, and the angles referred to the corresponding figure:

$$L = -(R_E + H_{OGS}) \sin \zeta + \sqrt{(R_E + H)^2 - (R_E + H_{OGS})^2 \cos^2 \zeta}$$

$$L = \sqrt{(R_E + H_{OGS})^2 + (R_E + H_{sat})^2 - 2(R_E + H_{OGS})(R_E + H_{sat}) \cos \gamma}$$

$$d_{max} = -R_E \sin \alpha + \sqrt{(R_E \sin \alpha)^2 + H^2 + 2 R_E H}$$

5 Pointing Error

5.1 Two-line element error

5.2 Star tracker error

5.3 Encoders reading error

5.4 Jitter

5.5 Atmospheric effects

6 Data Communication

6.1 Baud Rate

6.2 Bandwidth

6.3 Modulation

6.4 Spectral Efficiency

6.5 Capacity

$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

6.6 Signal-to-Noise Ratio

$$\mathrm{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \quad \rightarrow \quad 10 \cdot \log(\mathrm{SNR}_0) = 74.9 \text{ dB}$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\left\langle i_N^2 \right\rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

6.7 Bit Error Rate

$$\text{4-QAM: } BER \approx \cdot Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

$$\text{16-QAM: } BER \approx 3 \cdot Q \left(\sqrt{\frac{4E_b}{5N_0}} \right)$$

6.8 Mean SNR

$$\langle \mathrm{SNR} \rangle = \frac{\mathrm{SNR}_0}{\sqrt{\left(\frac{P_{S0}}{\langle P_S \rangle}\right) + \sigma_I^2(D_G)\mathrm{SNR}_0^2}}$$

$$\langle \mathrm{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \mathrm{SNR}_0 \to \infty$$

$$\langle \mathrm{SNR} \rangle = \frac{\langle i_S \rangle}{\sigma_N}$$

6.9 Fading statistics

$$Pr_{\mathrm{fade}} = 1 - \frac{1}{2}\int_0^\infty (u)\mathrm{erfc}\left(\frac{\mathrm{TNR} - \langle \mathrm{SNR} \rangle\, u}{\sqrt{2}}\right) du$$

$$Pr(E) = \langle \mathrm{BER} \rangle = \frac{1}{2}\int_0^\infty p_f(u)\mathrm{erfc}\left(\frac{\langle \mathrm{SNR} \rangle\, u}{2\sqrt{2}}\right) du$$

7 Modulation

7.0.1 OOK: On-Off Keying

7.0.2 QPSK: Quadrature Phase Keying

7.0.3 QAM: Quadrature Amplitude Modulation

7.1 Data Rates

8 Photodetectors

8.1 Noise

8.2 APD

8.3 PIN

8.3.1 InGaAS

Example: InGaAs PIN photodiode G6854-01

- Sensitivity S = 0.95 A/W
- Noise Equivalent Power: NEP = 2×10^{-15} W/$\sqrt{\mathrm{Hz}}$

8.4 QPD

9 Laser transmitter

9.1 Amplitude modulation

9.2 Phase modulation

9.3 Polarization modulation

10 References

?? https://www.nsnam.org/reviews/2016/socis-midterm/fso.html